Project: Unsupervised Learning for Pattern Discovery and Customer/User Segmentation

AdVantage Growth Studio

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1. Business Overview

AdVantage Growth Studio is a fictional data-driven marketing agency that serves clients in retail, e-commerce, and service industries across Canada. The agency helps clients improve campaign effectiveness by identifying which customers are most likely to respond to marketing outreach, and by tailoring strategies to different customer groups based on their behaviour and profile.

The organization has observed that marketing budgets are being wasted on broad, undifferentiated campaigns that fail to distinguish between customers with very different purchasing behaviours and engagement patterns. Clients are sending campaigns to all customers equally, resulting in low conversion rates and missed revenue opportunities from customers who might have responded with the right message or offer.

This assignment applies K-Means clustering to segment customers into natural groups based on their purchasing history, engagement behaviour, loyalty, satisfaction, and income profile. These segments enable AdVantage's clients to move from generic campaigns to targeted, data-driven marketing strategies.

Element	Description
Business	AdVantage Growth Studio
Objective	Segment customers into groups to support targeted marketing campaign strategies
Method	K-Means Clustering (unsupervised learning)
Dataset Size	605 customer records (600 after removing duplicates)
Features Used	9 numerical features
Number of Clusters	k=3 (selected via Elbow Method)
Algorithm	KMeans (sklearn, random_state=42, n_init=10)

2. Data Preparation

2.1. Dataset Overview

The dataset contains 605 records representing individual customers who received a marketing campaign from one of AdVantage's clients. Each record has 21 columns spanning six feature groups:

Feature Group	Columns
Customer Profile	Customer_ID, Campaign_Date, Age, Annual_Income, Customer_Segment
Campaign details	Campaign_Channel, Campaign_Type, Ad_Exposure_Count, Discount_Offered_Percent
Engagement behaviours	Email_Opened, Ad_Clicked, Website_Visits_Last_30_Days, Social_Media_Engagement_Score
Purchase history	Previous_Purchases, Average_Order_Value, Days_Since_Last_Purchase
Loyalty and satisfaction	Loyalty_Score, Customer_Satisfaction
Campaign outcomes	Prior_Campaign_Response, Converted

2.2. Preprocessing Steps

The following preprocessing steps were applied before clustering:

- Stripped extra whitespace from all column names to ensure clean column references
- Removed 5 duplicate rows (605 reduced to 600)
- Removed $ signs and commas from Annual_Income and Average_Order_Value and converted both columns from object to float
- Identified missing values: Annual_Income (10 missing), Average_Order_Value (6 missing), Customer_Satisfaction (8 missing), Customer_Segment (6 missing)
- Filled missing values in numerical columns with column medians
- Filled missing values in categorical columns with the column mode (most frequent value)
- Selected 9 numerical features for clustering: Age, Annual_Income, Website_Visits_Last_30_Days, Previous_Purchases, Average_Order_Value, Days_Since_Last_Purchase, Loyalty_Score, Customer_Satisfaction, Social_Media_Engagement_Score

Scaled all features using StandardScaler to ensure no single feature dominates due to differences in scale.

3. K-Means Clustering Model

3.1. How K-Means Clustering Works

K-Means is an unsupervised machine learning algorithm that groups data points into a fixed number of clusters based on similarity. Unlike supervised learning, K-Means does not use a target variable; it discovers natural patterns in the data on its own.

The algorithm follows five steps: (1) choose k, the number of clusters; (2) randomly initialize k cluster centres; (3) assign each data point to the nearest centre using Euclidean distance; (4) recalculate each centre as the mean of its assigned points; (5) repeat until centres no longer move. The result is a set of clusters where points within each cluster are as similar as possible to each other and as different as possible from other clusters.

3.2. Elbow Method Analysis

To choose the right value of k, the Elbow Method was applied. K-Means was trained for k=1 through k=10. For each value of k, the inertia (total within-cluster sum of squared distances) was recorded. The optimal k is the point where reducing k further yields rapidly diminishing returns, marking the bend or elbow in the curve.

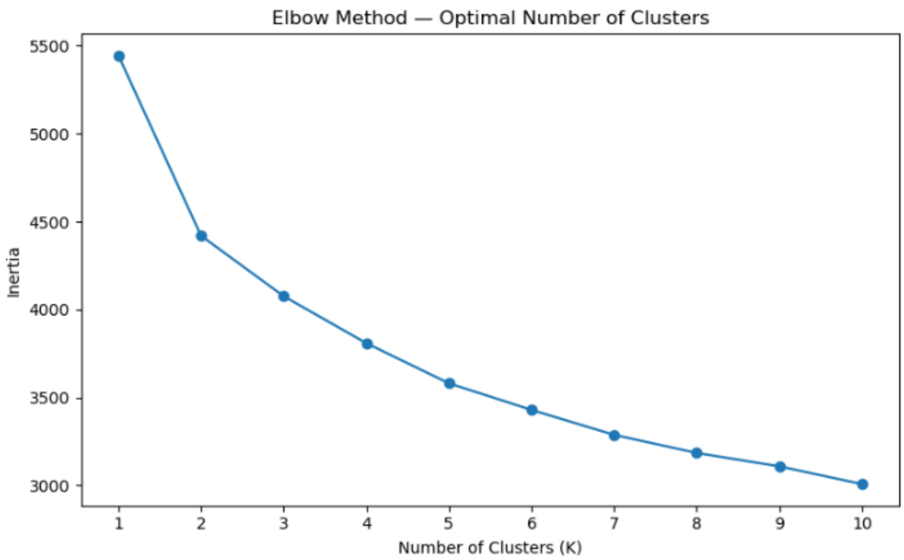

Figure 1: Elbow Method, AdVantage Growth Studio

k	Inertia	k	Inertia
1	5,445.00	6	3,429.42
2	4,419.77	7	3,287.53
3	4,076.04	8	3,184.61
4	3,808.99	9	3,108.53
5	3,580.75	10	3,007.64

Table 1: Inertia Values for k=1 to k=10

The curve drops steeply from k=1 to k=3, after which it begins to flatten. The most significant drops occur between k=1 and k=2 (approximately 1,026 points) and between k=2 and k=3 (approximately 343 points). After k=3, the reductions become progressively smaller and relatively consistent. The elbow at k=3 represents the best balance between cluster compactness and model simplicity. Although k=4 or k=5 could be considered because the elbow is not extremely sharp, k=3 provides the best balance between reducing inertia and maintaining model simplicity. This aligns with the business objective of creating customer segments that are both statistically meaningful and operationally useful.

4. Cluster Analysis and Interpretation

4.1. Cluster Summary Statistics

After training the K-Means model with k=3 and random_state=42, cluster labels were added back to the dataset. The table below shows the mean value of each feature per cluster.

Feature	Cluster 0 (191)	Cluster 1 (227)	Cluster 2 (187)
Age	35.78	40.08	36.83
Annual Income ($)	$65,690.54	$50,515.53	$72,626.53
Website Visits (Last 30 Days)	6.40	6.04	6.14
Previous Purchases	6.74	1.80	1.83
Average Order Value ($)	$84.13	$68.87	$107.44
Days Since Last Purchase	22.29	49.46	72.52
Loyalty Score	75.50	41.84	35.79
Customer Satisfaction	7.56	6.27	5.66
Social Media Engagement Score	42.20	51.01	41.27
Cluster Size	**191 customers**	**227 customers**	**187 customers**

4.2. Cluster Interpretation

4.2.1. How many customers/user segments did you find?

The K-means clustering model identified three customer segments (K = 3). The value of K was selected using the Elbow Method, which indicated that three clusters provided the best balance between model complexity and explanatory power. Cluster 1 is the largest segment, while Cluster 2 is the smallest.

The cluster distribution is shown below:

Cluster	Segment Name	Number of Customers
Cluster 0	Loyal Active Customers	191
Cluster 1	Low-Loyalty Social Browsers	227
Cluster 2	High-Income Dormant Customers	187

4.2.2. Main Characteristics of Each Segment

4.2.2.1. Cluster 0: Loyal Active Customers (191 Customers)

These customers are the most engaged and loyal in the dataset. They record the highest loyalty score (75.50), the highest previous purchases (6.74), and the highest customer satisfaction (7.56). They have also purchased most recently, with an average of only 22.29 days since their last purchase. Their average annual income is $65,690.54 and their average order value is $84.13. Although not the largest segment, these customers generate disproportionate value through frequent repeat purchases and strong satisfaction levels. This group forms the foundation of the business's recurring revenue and should be the primary focus of retention efforts.

4.2.2.2. Cluster 1: Low-Loyalty Social Browsers (227 Customers)

This is the largest segment at 227 customers, representing approximately 37.5% of the total customer base. These customers record the highest social media engagement score (51.01) but have made very few purchases, averaging only 1.80 previous purchases and the lowest average order value ($68.87). Their loyalty score is moderate (41.84), and they have not purchased for an average of 49.46 days. Their annual income is the lowest of the three segments ($50,515.53). The pattern suggests that marketing reach and digital exposure are not translating into purchasing behaviour for this group. This segment is the primary target for conversion-focused interventions.

4.2.2.3. Cluster 2: High-Income Dormant Customers (187 Customers)

These customers have the highest average annual income ($72,626.53) and the highest average order value ($107.44), yet they have not purchased for the longest time, averaging 72.52 days since their last purchase. Their loyalty score is the lowest of all three clusters (35.79), and customer satisfaction is also the lowest (5.66). They record the fewest previous purchases (1.83) despite their high spending power. Although currently inactive, their high order values make them a high-priority reactivation target, as even a small number of re-engaged customers could generate significant revenue.

4.2.3. Difference between each segment

Days Since Last Purchase: The strongest differentiating variable across the clusters is Days Since Last Purchase.

- Cluster 0: ~22 days (actively purchasing)
- Cluster 1: ~50 days (occasional or lapsing customers)
- Cluster 2: ~73 days (clearly dormant customers)

This measure clearly separates active customers from inactive ones.

Loyalty Score: Loyalty score strongly distinguishes Cluster 0 from the other two segments.

- Cluster 0: 75.5
- Cluster 1: ~41.8
- Cluster 2: ~35.8

However, the reasons for low loyalty differ. Cluster 1 customers have not purchased frequently enough to develop loyalty, whereas Cluster 2 customers appear to have disengaged despite having a stronger purchasing history.

Social Media Engagement: Social media engagement reveals an interesting pattern.

- Cluster 1: 51.0 (highest)
- Cluster 0: ~42.2
- Cluster 2: ~41.3 (lowest)

The segment that purchases the least is actually the most active online. This suggests that visibility and engagement alone do not necessarily translate into purchases.

Income vs. Spending Behaviour: Cluster 0 and Cluster 2 have very similar annual incomes:

- Cluster 0: ~$65,700
- Cluster 2: ~$72,600

Despite similar earning power, their behaviours differ dramatically. Income alone does not explain customer value; recency, loyalty, and engagement are much stronger indicators of purchasing behaviour.

Segment Size: Segment size is another important distinction.

- Cluster 1: 227 customers
- Cluster 0: 191 customers
- Cluster 2: 187 customers

Although Cluster 1 is the largest segment, it generates the fewest purchases per customer. This means the business's largest customer group is currently its least productive from a revenue perspective.

4.2.4. Most Valuable Segment

Cluster 0 is the most immediately valuable customer segment. These customers:

- Purchase most frequently

- Have the highest loyalty score (75.5)
- Have the highest satisfaction levels (~7.6)
- Maintain a strong purchase history (~7 previous purchases)

Although they are not the largest segment, they provide a dependable source of repeat revenue and require relatively low acquisition costs. Retaining these customers through loyalty rewards, personalized promotions, and exclusive offers would likely generate the highest return on investment.

4.2.5. Segment Requiring More Marketing Attention

Cluster 1 requires the most urgent marketing attention. There are two primary reasons for this:

a) **It is the largest segment in the customer base:** With 227 customers (approximately 37.5% of all customers), Cluster 1 represents a substantial proportion of the market. However, these customers average only **1.82 previous purchases**, indicating very poor conversion performance.

b) **They are highly engaged but rarely purchase:** The segment records the highest social media engagement score (**51.0**), demonstrating that marketing efforts are successfully reaching them. The problem is not awareness, it is conversion.

This suggests that marketing strategies should focus on:

- First-purchase incentives
- Stronger calls-to-action
- Retargeting campaigns
- Promotional discounts
- Simplified purchasing journeys
- Conversion-focused messaging

Without targeted intervention, this segment is likely to continue consuming marketing resources while generating limited revenue.

5. Cluster Visualizations

5.1. Elbow Method Plot

The Elbow Method plot shows inertia decreasing as the number of clusters increases. The steepest drops occur between k=1 and k=3. After k=3, the reduction in inertia becomes progressively smaller, indicating diminishing analytical value from additional clusters. The selected value of k=3 is marked on the plot with a red dashed vertical line.

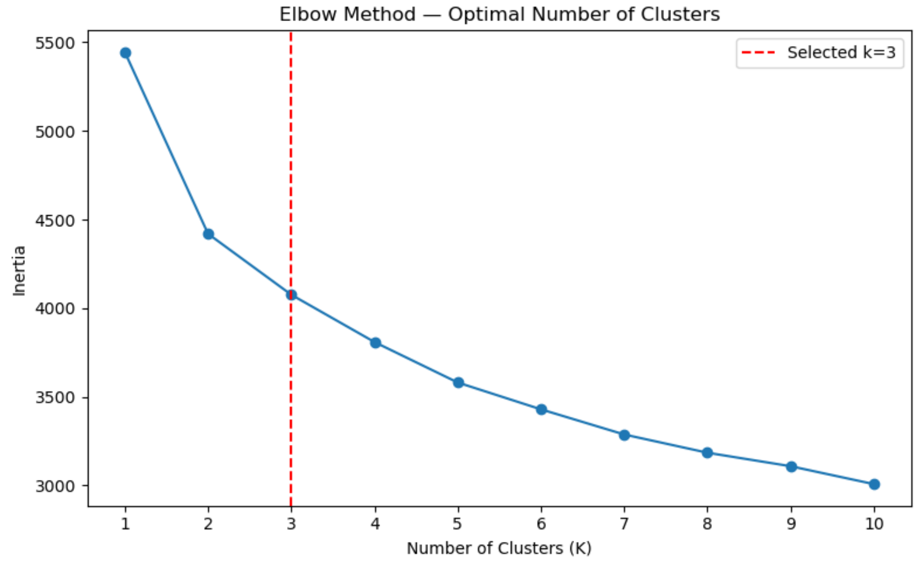

Figure 2: Elbow Method Plot with selected k=3 marked

5.2. Scatter Plot

The three clusters separate clearly in two-dimensional space.

- **Cluster 0 (blue)** sits in the mid-income, high-loyalty zone. These are the most engaged customers.
- **Cluster 2 (orange)** sits in the high-income, mid-loyalty upper band. They have spending power but aren't currently active.
- **Cluster 1 (green)** occupies the low-income, low-loyalty area. This group shows heavy social-media exposure but minimal purchasing intent.

This clear separation between clusters validates that K-means found genuinely distinct behavioural groups.

Figure 3: Scatter Plot of Annual Income vs Loyalty Score by cluster

5.3. Bar Chart

Each group of bars shows the average value of a key feature broken down by cluster. This makes it easy to spot where the three segments differ most.

a) **Loyalty Score:** Cluster 0 (Loyal Active) leads clearly with a loyalty score of 75.5, while Cluster 1 (Low-Loyalty Browsers) and Cluster 2 (High-Income Dormant) are much closer together at 41.8 and 35.8, respectively. This indicates that both non-loyal groups exhibit similarly weak loyalty, with Dormant customers scoring slightly lower than Browsers.

b) **Annual Income (÷1000):** Cluster 2 (High-Income Dormant) lives up to its name with an average income of 65.7, although Cluster 0 is not far behind at 65.7. Cluster 1 trails at 50.5. This suggests that income alone does not distinguish Clusters 0 and 2; instead, their behaviours and engagement patterns are what separate them.

c) **Days Since Last Purchase:** Cluster 2 stands out sharply with an average of 72.5 days since the last purchase, which is more than three times longer than Cluster 0's 22.3 days. Cluster 1 falls in between at approximately 49.5 days. This is the strongest behavioural differentiator among the clusters, indicating that Cluster 2 customers have been inactive for the longest period.

d) **Social Media Engagement:** Interestingly, Cluster 1 (Browsers) records the highest social media engagement score at 51.0, followed by Cluster 0 (Loyal Active) at 42.2, while Cluster 2 (Dormant) has the lowest score at 41.3. This finding is somewhat counterintuitive, as the least active purchasers are the most active on social media.

e) **Average Order Value (÷10):** Cluster 2 again ranks highest with an average order value of 10.7, compared with 8.4 for Cluster 0 and 6.9 for Cluster 1. Although these customers purchase less frequently, they tend to spend more when they do make a purchase, making them a particularly valuable target for reactivation campaigns.

f) **Customer Satisfaction:** Customer satisfaction scores are relatively similar across all three clusters, with Cluster 0 scoring 7.6, Cluster 1 scoring 6.3, and Cluster 2 scoring 5.7. Satisfaction therefore does not strongly differentiate the segments. However, the slightly lower satisfaction level reported by Dormant customers may provide some insight into their disengagement.

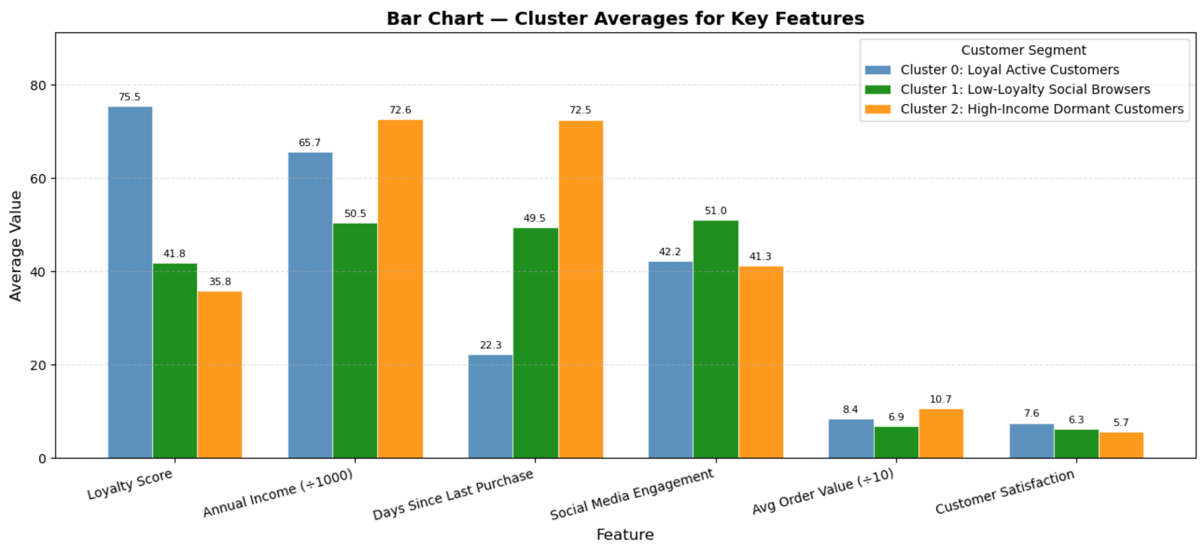

Figure 4: Bar chart comparing cluster averages across key features

5.4. Box Plot

The box plot panel shows the distribution of five features across the three clusters: Loyalty Score, Days Since Last Purchase, Social Media Engagement Score, Customer

Satisfaction, and Previous Purchases. The x-axis of each subplot displays the three cluster groups (C0, C1, C2) with colours matching the scatter plot.

a) **Loyalty Score**: Cluster 0 (blue) displays a high median loyalty score (approximately 75) and a relatively compact box, indicating that loyal customers are consistently engaged. In contrast, Cluster 1 (green) and Cluster 2 (orange) have much lower medians, around 40 and 35 respectively, with wider distributions. This suggests greater variability in customer loyalty within these groups and confirms that consistently high loyalty is a defining characteristic of Cluster 0.

b) **Days Since Last Purchase:** Cluster 2 (orange) exhibits the highest and most dispersed distribution, with a median of approximately 70–80 days since the last purchase and a large interquartile range extending beyond 140 days. Several outliers are also visible above 120 days, indicating highly inactive customers. By comparison, Cluster 0 (blue) is tightly concentrated around a median of 20 days, reflecting recent and consistent purchasing activity. Cluster 1 (green) falls between the two, with a median near 50 days and moderate variability.

c) **Social Media Engagement Score:** The distributions for social media engagement overlap considerably across all three clusters, with median values ranging from approximately 40 to 65. Cluster 2 (orange) has the highest median and the widest spread, indicating substantial variation in engagement levels. Some Dormant customers are highly active on social media, while others show limited engagement. This variability suggests that broad social media campaigns may be less effective for Cluster 2 without further customer segmentation.

d) **Customer Satisfaction:** Cluster 0 (blue) demonstrates the highest customer satisfaction, with a median score of approximately 8–9 and a relatively narrow distribution. This indicates that loyal customers are generally satisfied and share similar experiences. Clusters

1 and 2 both have lower median satisfaction scores, around 5–6, and wider ranges. Notably, Cluster 2 (orange) extends to particularly low satisfaction values, supporting the possibility that dissatisfaction may be a contributing factor to customer disengagement.

e) **Previous Purchases:** Cluster 0 (blue) records the highest median number of previous purchases (approximately 7) with a moderately compact distribution, reflecting a strong purchasing history. Cluster 1 (green) has the lowest median, around 2, and a relatively narrow spread, indicating limited prior purchasing activity. Cluster 2 (orange) occupies the middle ground with a median of approximately 5–6, but displays a wider distribution. This suggests that while some Dormant customers were previously frequent buyers, others had relatively low purchasing activity even before becoming inactive.

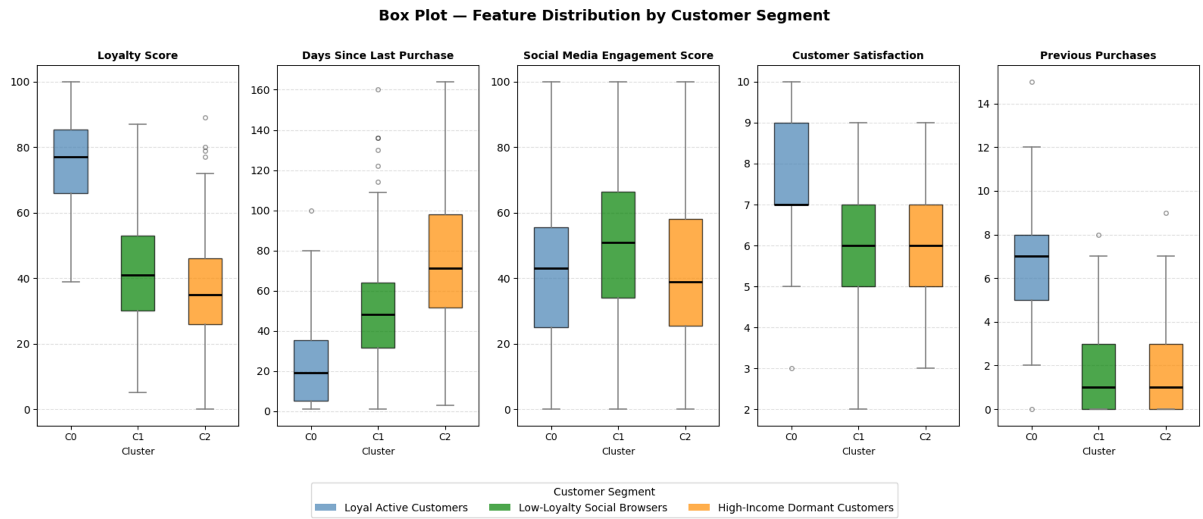

Figure 5: Box plot panel showing feature distributions by customer cluster

6. Business Actions and Recommendations

Based on the clustering results, the following marketing strategies are recommended for each customer segment:

Cluster	Insight	Recommended Marketing Action
Cluster 0: Loyal Active Customers	High loyalty, frequent purchases, strong satisfaction	Implement a formal loyalty rewards programme. Provide early access to new products or exclusive member-only offers. Use purchase history to recommend complementary products.

		Send personalized appreciation messages to reinforce the relationship.
Cluster 1: Low-Loyalty Social Browsers	High social media engagement but low conversion and few purchases	Run retargeting campaigns on social media platforms. Offer first-purchase discounts or free-shipping incentives. Use social proof such as customer reviews and testimonials. Test urgency-based and value-based messaging. Simplify the checkout process to reduce purchase friction.
Cluster 2: High-Income Dormant Customers	High income and order value but long inactivity and low satisfaction	Launch personalized win-back email campaigns with premium incentives. Recommend products based on previous purchase history. Conduct short satisfaction surveys to identify causes of disengagement. Offer VIP-style service or exclusive benefits to reactivate high-value accounts.
All clusters: Replace informal judgment	Use cluster membership as a data-driven early warning signal	Assign cluster labels within the CRM system. Route each segment into separate marketing workflows. Monitor cluster-level KPIs quarterly. Re-run the clustering model periodically with updated data to ensure segment accuracy.

7. Limitations

The following limitations should be considered before applying these results in a real marketing environment:

- **Static snapshot:** The dataset represents customer behaviour at a single point in time. Customer behaviour is dynamic and can change significantly. A Cluster 1 browser could become a Cluster 0 loyal customer after a successful campaign. Cluster labels should be refreshed periodically using updated data.
- **K-Means assumptions**: K-Means assumes clusters are roughly spherical and similar in size. Customer groups rarely form perfectly round or evenly distributed patterns, and the algorithm is sensitive to the initial random placement of cluster centres. Different random seeds could produce slightly different cluster assignments.

- **Synthetic data:** The dataset was synthetically generated for educational purposes. Real customer data would likely produce different patterns and would require additional validation, domain knowledge, and regulatory compliance considerations before acting on results.
- **No causal inference**: Clustering reveals correlation patterns, not causal relationships. The fact that Cluster 1 has high social media engagement and low purchases does not prove that social media exposure causes poor conversion. Other factors, such as price sensitivity, trust, or purchasing intent, may be at play.
- **Predefined k:** The number of clusters must be chosen before running the algorithm. While the Elbow Method suggested k=3, this selection involves subjective judgment. Choosing k=4 or k=5 may reveal additional sub-segments that are currently combined within a single cluster.

8. Responsible AI Considerations

Marketing analytics applications that segment customers can directly influence which customers receive promotional benefits, discounts, and outreach. The following responsible AI principles apply to this analysis:

- **Fairness:** Cluster membership should never be used to make blanket exclusionary decisions about individuals. Cluster 1 contains customers with the lowest average income; decisions that consistently deprioritize this segment could unintentionally disadvantage lower-income customers. Income often correlates with demographic factors, so cluster-based decisions should be reviewed for unintended discriminatory effects.
- **Human oversight:** Clustering results should always be reviewed by marketing managers and business stakeholders before any campaign actions are taken. The model identifies patterns; human judgment must determine the appropriate response for each situation and each customer.

- **Transparency:** Customers whose data is used in analytics should be informed that their information is being used to support marketing planning and personalization. Organizations have a responsibility to be transparent about how data-driven tools influence customer treatment and campaign decisions.
- **Privacy:** All customer data used in this analysis must be stored and processed in accordance with applicable privacy legislation, including Canada's Personal Information Protection and Electronic Documents Act (PIPEDA). Personal identifiers such as Customer_ID and Campaign_Date should not be shared beyond authorized marketing and analytics teams.